Lab 1

Part I: 80 points

1.) In class I defined an orthonormal transformation matrix (see lecture 2)
    a. Use Claude to write a program that does the following:
        i. Use python and tkinter as in my example project. Follow the required pattern, requirements_v1, requirements_v2, etc.
        ii. User specifies a transform matrix size (restrict valid values to a power of 2, e.g., 2, 4, 8, etc. Make sure program checks for this constraint and notifies if it is violated)
        iii. Claude generates the orthonormal matrix I specified in my lecture
        iv. A “verify” button that, when pressed, multiplies the matrix by its transpose, verifies it is the identity matrix, and displays its result in a log window. Does this prove the rows are orthonormal? Why or why not? Justify your answer
        v. You specify a row number, press a button, and claude generates a graph of that row of the matrix (this is graphing the basis vectors of the transform)

vi. A menu where you can pick an excel spreadsheet file as the input. A toggle that allows you to select forward or inverse transform. A field where you can select the column of the spreadsheet to process, and a field where you specify the column of the spreadsheet in which to place the processed result. The spreadsheet will have a vector size (call it N) in the first row of its first column (e.g. 2, 4, 8, 16, etc.). rows 1 through N of columns 2, 3, 4, etc. are "data" columns and can hold the components of a vector to be transformed (or inverse transformed), or the result of such a transformation. A "go" button which will read the selected spreadsheet, extract the data values from the selected column, transform (or inverse transform) the values in that column and put the result in the specified column and graph the result

vii. A button and field where you pick a column of the spreadsheet to graph, and the graph is generated when the button is pressed. It should also compute the energy of the selected vector

b. Now, use your program to look at the transform of various data vectors. Of course, you're going to need to do a bit of excel programming to generate these data vectors. Set the transform size to 128

i. What is the transform of cos(2*pi*6*n/128)? (describe your result in words).
ii. Inverse transform the result. Do you get the original data column back?
iii. What is the transform of sin(2*pi*6*n/128)? Explain your result and verify the inverse transform performs as expected
iv. What is the transform of cos(2*pi*9*n/128)? Explain and verify as before
v. What is the transform of sin(2*pi*9*n/128)? Explain and verify as before
vi. What is the transform of the sum of the previous two vectors? Explain and verify as before
vii. What is the transform of cos(2*pi*6*n/128 + theta), where theta corresponds to a 45 degree phase shift (you will need to convert it to radians)? What is this telling you about this sinusoid? Is there a trigonometric identity which relates to this result?
viii. Generate a vector with random values uniformly distributed between -1 and 1 and take its transform. Graph the input and graph the transform. What are the energies of the input vector and its transform?
ix. Generate a pulse of width 10 (this is 10 1's followed by zeroes). Graph the pulse and

transform. What are the energies of the the pulse and transform?

x. Generate a pulse of width 5 and graph the transform. What are the energies of the pulse and transform?

xi. Generate a pulse of width 30 and graph the transform. What are the energies of the pulse and transform?

xii. Explain what you see in parts ix, x, xi, and xii